

Encoding a Qubit on a Torus

PH5842 Course Presentation

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Primary refs: [Dennis2002], [Bluvstein2022]

Introducing Toric Code

- Physical qubits reside on every edge (aka “link”) of a $L \times L$ square lattice
- To make it a Torus, consider top/bottom and right/left edges wrapped around respectively (so L sites and L links in each direction respectively)

We define the “check” operators for this lattice:

Plaquette: $Z_P = \bigotimes_{\ell \in P} Z_\ell$

Site: $X_s = \bigotimes_{\ell \ni s} X_\ell$

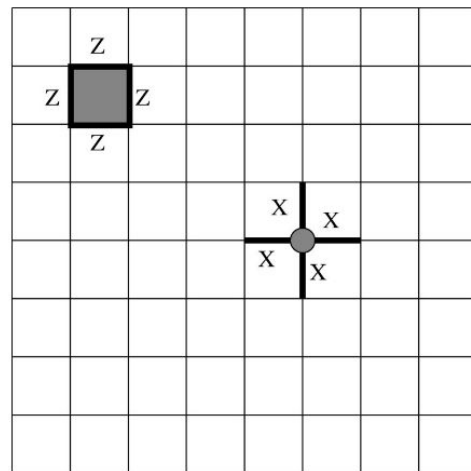


Fig 1: Check Operators on a square grid ₂

Reference: [Dennis2002]

Check operators

- $L \times L$ Plaquette and Site operators each
- However, product of all check operators of one type = 1, so there exist only $L^2 - 1$ unique check operator of each type
- From the independent set, every check operator commutes with every other, hence can form generators of an Abelian group
- Toric code: +1 eigenstate of the set of all check operators (stabilizer formalism)
- No of encoded qubits = $2L^2 - 2(L^2 - 1) = 2$

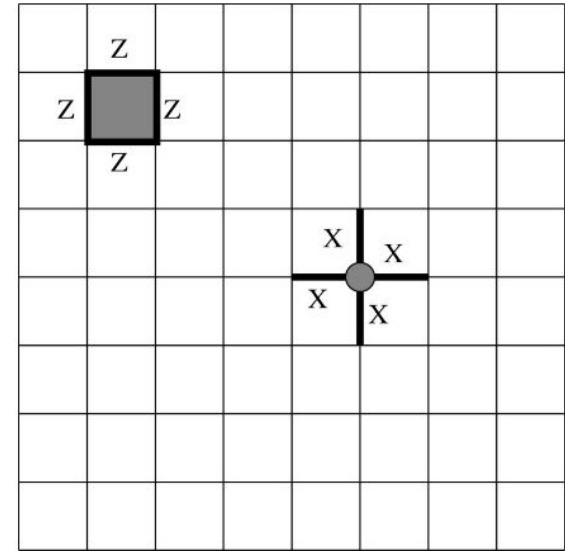


Fig 1: Check Operators on a square grid

Normalizer set

Any n-qubit Pauli operator: expanded into tensor prod of $\{I, X\}$ s and $\{I, Z\}$ s
(consider just Zs for now)

For the tensor prod. of Zs to commute with the entire stabilizer set, every site must have **2 Z links attached**; only possible if they occur in links that form a closed cycle
(Similar argument holds for X but in Dual lattice)

Therefore, Any general Pauli in the Normalizer: broken down into,

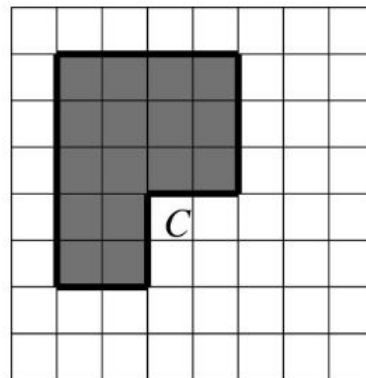
$$(\text{Cycle of Z links in lattice})^{\otimes} (\text{Cycle of X links in Dual lattice})$$

Identifying Logical Operations

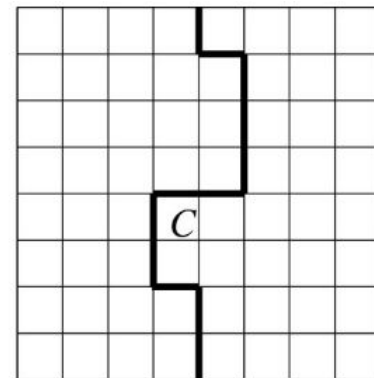
Fig 2.

Two types of Cycles on the Lattice:

- a) Can be written as a product of Plaquettes
- b) Cannot be represented as a product of plaquettes (but still a cycle because Torus wraps around)



(a)



(b)

Both commute with Stabilizer, but (b) does NOT lie in it $\Rightarrow$ (b) lies in the normalizer (preserves codespace, but not codeword)

(Gives intuition for why distance of the code = L) (!!!)

Identifying Logical Operations

Z, **X** (defined as in Fig 3) both commute with codespace stabilizers.

- a. $\mathbf{Z.Z} = \mathbf{X.X} = \mathbf{I}$
- b. $[\mathbf{Z}, \mathbf{X}] = -[\mathbf{Z}, \mathbf{X}]$ (overlap at 1 qubit)
- c. $[\mathbf{Z}, \mathbf{Z}] = [\mathbf{X}, \mathbf{X}] = 0,$

i.e. we can define **Z**, **X** to form our Logical Pauli operators

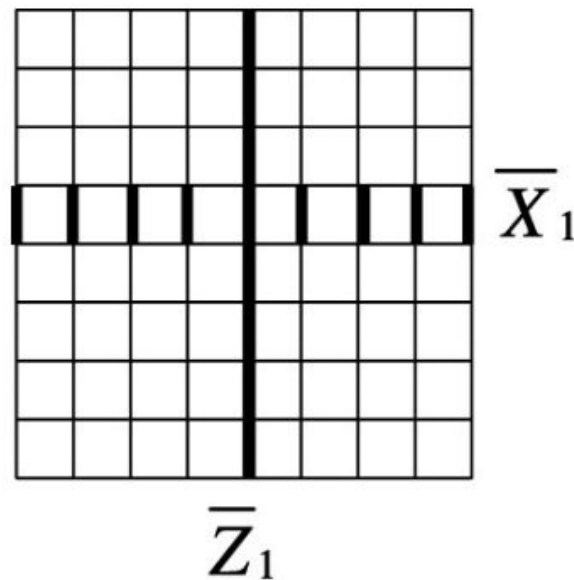


Fig 3. Logical Codespace operators

Errors on the Lattice

Error “defect”: A check measurement returned -1

Can localise this to a lattice point that defines the plaquette (site)

Each qubit has two stabilizer generators of one kind

=> Error on one qubit manifests in two measurements (i.e., causes 2 defects on the lattice)

Error on a string of such links => Appears at ends of the chain!

For recovery, we just have to apply Zs (Xs) along a chain connecting the defects (**that loops around the right way**)

Reference: [Dennis2002]

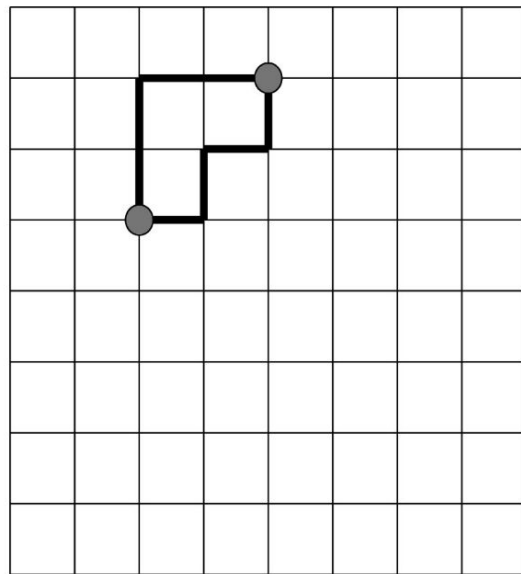


Fig 4: Two possible cycles, given defects on a lattice

What makes the code effective

Take error probability on each qubit as p

The number of expected errors $\sim O(pL^2) \gg L/2$ (which is no. of correctable errors $\sim d/2$)

$d=L$: misleading; for large L (and small enough p), we would need long connected links to affect our codespace (highly atypical)

Threshold $p \sim 0.11$ (assuming perfect detection; analysis in case of measurement errors, non trivial)

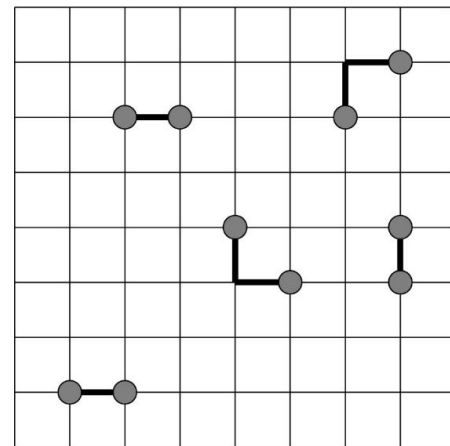


Fig 5: Errors occur in pairs and appear diluted in a lattice with large L (and small p)

Preparing the encoded state

Assuming perfect measurements, the following can be used to encode qubits in logical state 0:

- Initialize every qubit to 0 (state $|000\dots\rangle$)
- It is a simultaneous eigstate of Plaquettes but not sites
- Measure all site operators. About half the site operators given +1, about half give -1 (each state can be written as $|0\rangle^{\otimes 4} = (|+\rangle + |-\rangle)^{\otimes 4}$)
- Each “-1” is hereby considered a defect and corrected for by syndrome recovery, thus bringing everything to +1 eigstate

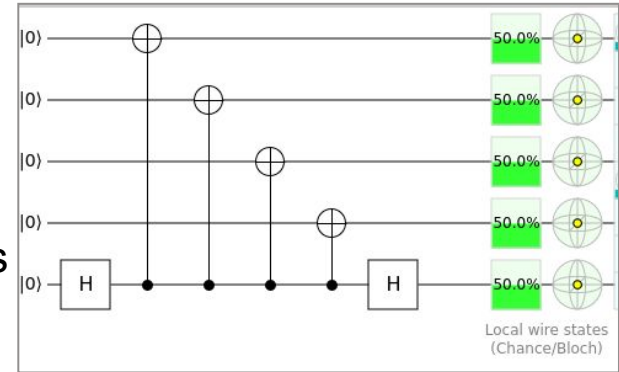


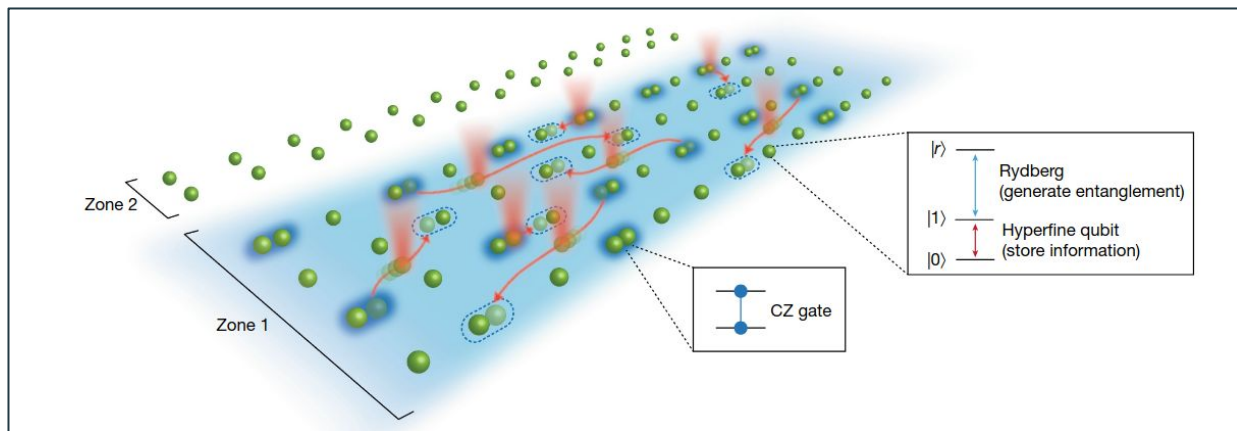
Fig 6: Measuring site operators using ancilla (X1X2X3X4 measurement)

Implementing Toric code with Atomic qubits

Neutral atom (Rb87) QC Architecture:

Non-local connectivity enabled via dynamic traps

Gates that can be performed: Ry (global) and Z (local) (High fidelity, $\sim 99.9\%$), CZ ($\sim 97.5\%$)

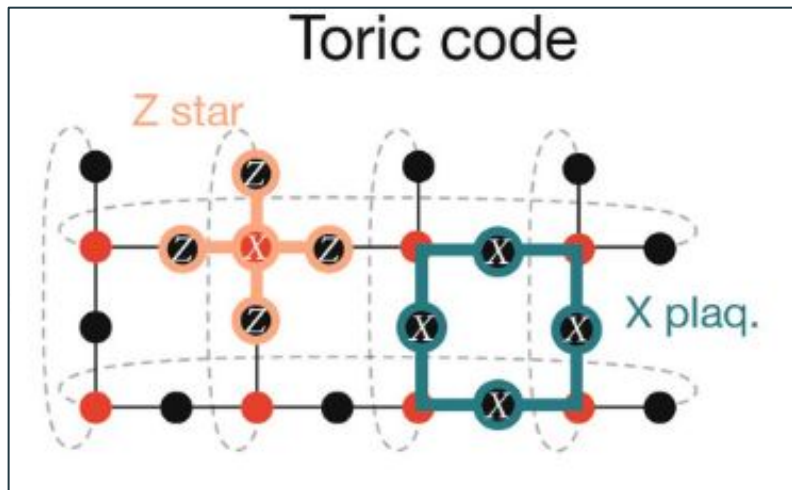


Fidelity ~ 1 for transport $< 200\mu\text{m}$

Fig 7: Non-local connectivity in a neutral atom architecture

Reference: [Bluvstein2022]

d=4 Toric code



- 16 block qubits, 8 ancillae
- 7 independent plaquettes/sites
- QEC round: $1\text{ms} \ll T_2 (\sim 1\text{s})$
- Direct CZ between physical qubits is difficult so ancilla-assisted state preparation

Fig. 8 Stabilizers for the d=4 code. Black are data, red are ancilla. Total of 16 data and 8 ancilla qubits.

Encoding to logical 0 in practice

[Video link](#)

Toric code state $|+\rangle|+\rangle$ is prepared (corresponding to topologically distinct directions)

Small diff. in implementation from earlier discussion:

1. Post-measurement, rather than sites being corrected the stabilizers are updated classically
2. ancillas are moved in space (using dynamic traps) to perform CZ gates with each of their four neighbours in one zone, then moved to a separate zone and measured (in a different basis)

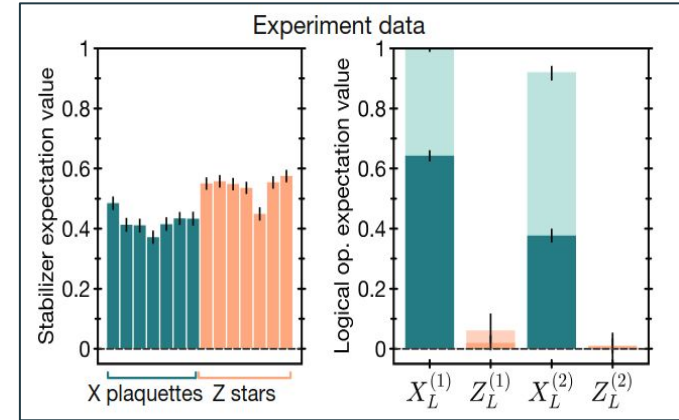


Fig 9: Verifying state preparation. $\langle X1 \rangle = 0.64$, $\langle X2 \rangle = 0.38$

$\langle X \rangle = 1$ when conditioned on datasets with no syndrome detection

Comments on the Toric code

Advantages:

- Easy to implement on Atom-based hardware
- Offers a universal set of transversal gates (not covered here)

Disadvantages:

- Difficult to have “concatenated” codes, again due to architecture complexity.
(Required for Robust quantum memory applications)
- Planar might be better suited across other hardware architectures due to local connectivity.

References

[Bluvstein2022]: Bluvstein, D., Levine, H., Semeghini, G. *et al.* A quantum processor based on coherent transport of entangled atom arrays. *Nature* **604**, 451–456 (2022).

[Dennis2002]: Eric Dennis, Alexei Kitaev, Andrew Landahl, John Preskill; Topological quantum memory. *J. Math. Phys.* 1 September 2002; 43 (9): 4452–4505.

Fowler et. al.; Surface codes: Towards practical large-scale quantum computation. *Phys. Rev. A* **86**, 032324

Quirk: <https://algassert.com/quirk>